

7

NUMBER AND ALGEBRA

EQUATIONS

Equation-solving has been recorded as far back as 1500 BCE. It was first used in ancient Babylon and Egypt and was brought to Europe from India by the Arabs during the 9th century. The word 'algebra' comes from the Arabic word *al-jabr*, meaning restoration, the process of performing the same operation on both sides of an equation to solve the equation.



Chapter outline

	Working mathematically				
7.01 Two-step equations	U	F		R	
7.02 Equations with variables on both sides	U	F		R	
7.03 Equations with brackets	U	F		R	C
7.04 Equation problems	U	F	PS	R	C
7.05 Simple quadratic equations $ax^2 = c$	U	F		R	C
7.06 Equations and formulas	U	F	PS	R	

Wordbank

Quiz
Wordbank 7

equation A mathematical statement that two quantities are equal, involving algebraic expressions and an equals sign (=)

formula A rule written as an algebraic equation, using variables

inverse operation An opposite used in solving an equation, for example, the inverse operation of multiplying is dividing

linear equation An equation involving a variable that is not raised to a power, such as $2x + 9 = 17$.

quadratic equation An equation involving a variable squared (power of 2), such as $3x^2 - 6 = 69$.

solution The answer to an equation or problem, the correct value(s) of the variable that makes an equation true

solve (an equation) To find the value of an unknown variable in an equation

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

7.01 Two-step equations 1 • Two-step equations 2
7.02 Equations with variables on both sides
7.03 Solving equations • Equations with brackets
7.05 Simple quadratic equations $ax^2 = c$
7.06 Equations with formulas 1 • Equations with formulas 2

***Twig* video (1):**

7.01 The Arabic science of balancing

***PhET* interactives (2):**

7.01 Equality explorer: Basics • Equality explorer

Quizzes (6):

- Wordbank 7
- SkillCheck 7
- Mental skills 7A
- Mental skills 7B
- Language of maths 7
- Test yourself 7

Skillsheets (3):

7.01 Solving equations by balancing • Solving equations by backtracking • Solving equations using diagrams

Worksheets (6):

7.03 Equations 2 • Checking solutions
7.04 Word problems with equations • Angle problems with algebra
7.06 Working with formulas
Mind map Equations

Puzzles (7):

7.01 Solving equations • Backtracking
7.02 Equations with unknowns on both sides
7.03 Equations • Equations order activity
7.04 Writing and solving equations
Language of maths Equations crossword

Presentation (1):

7.04 Applying linear equations

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve linear equations including those involving one algebraic fraction
- solve simple quadratic equations of the form $ax^2 = c$
- use equations and formulas to solve problems

Quiz
SkillCheck 7

SkillCheck ANSWERS ON P. 579

1 Use $x = -4$ to evaluate each expression.

a $5x$ **b** $x + 10$ **c** $x - 1$ **d** x^2
e $15 - x$ **f** $3(x - 1)$ **g** $7(x + 5)$ **h** $\frac{x}{2} + 3$

2 What operation is the opposite to:

a adding? **b** dividing? **c** multiplying? **d** subtracting?

3 Simplify each expression.

a $6n - 2n$ **b** $8x + 3x$ **c** $-r - 4r$ **d** $-8t + 2t$

4 Expand each expression.

a $4(2k + 7)$ **b** $5(3d - 1)$ **c** $-3(1 + 2y)$
d $-4(b - 8)$ **e** $5(3d + 2) + 8$ **f** $2(a + 5) + 5(a + 2)$
g $4(1 + d) - 3(1 - 5d)$

5 Solve each equation. Use substitution to check your solutions.

a	$m + 10 = 16$	**b**	$5a = 45$	**c**	$y - 6 = 2$
d	$\frac{w}{5} = 10$	**e**	$-3y = 27$	**f**	$27 = m + 4$
g	$6a = -42$	**h**	$\frac{a}{3} = 4$	**i**	$\frac{w}{-5} = 6$

6 Using n to represent 'the number', write an expression for each of these statements.

a The product of the number and 7.

b The square of the number.

c 5 times the sum of the number and 8.

d The number decreased by 20.

e The product of 6 and the number, decreased by nine.

f If the number is even, the *next* even number.

Two-step equations

7.01

An **equation** is a statement involving a variable (such as x), numbers and an equals (=) sign. If the variable is not raised to a power, then the equation is a **linear equation**. For example, $2x - 5 = 11$ is a linear equation, but $x^2 + 8 = 24$ is not because the x is squared (raised to the power of 2).

To **solve** an equation, we find the value of the variable that makes the equation true. This value is called the **solution** of the equation.

There are two algebraic methods for solving equations.

The **balancing method** involves representing both sides of an equation as balance scales, and solving the equation by performing the same operation on both sides to keep it 'balanced'.



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The **backtracking method** involves using a flowchart to show what operations are performed on the variable (say, x) to create the equation, then using a reverse flowchart to undo each operation by performing the inverse (opposite) operation in reverse order.



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Operation	Inverse operation
+	−
−	+
×	÷
÷	×

Puzzles
Solving equations

Backtracking

Skillsheets
Solving equations by balancing

Solving equations by backtracking

Solving equations using diagrams

Videos
The Arabic science of balancing

Two-step equations

Interactives
Equality explorer: Basics

Equality explorer

To **solve an equation**, aim to have the variable (such as x) on one side of the equation and a number on the other side, in the form:

$$x = ____$$

Check your solution by substituting it back into the equation.

Two-step equations require 2 steps or operations to solve.

Example 1

Solve each equation.

a $5x + 1 = 11$ **b** $\frac{5m}{2} = -3$ **c** $\frac{w}{3} - 6 = 5$

Solution

a **Method 1: The balancing method**

$5x + 1 = 11$

$5x + 1 - 1 = 11 - 1$ *Step 1:* Subtracting 1 from both sides

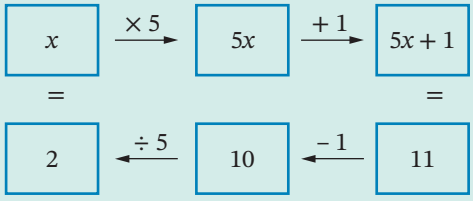
$5x = 10$

$\frac{5x}{5} = \frac{10}{5}$ *Step 2:* Dividing both sides by 5

$x = 2$ Check: $5 \times 2 + 1 = 11$

Method 2: The backtracking method

First consider how we get from x to $5x + 1$ using a flowchart.



In this equation, $5x + 1 = 11$. To get back to x, use the 'reverse flowchart' to undo the operations 'multiply by 5' and 'add 1' in the **reverse order**.

Subtract 1 and divide by 5 to find the solution $x = 2$.

Using algebra, we have:

$5x + 1 = 11$

$5x = 11 - 1$ *Step 1:* Undo '+ 1' by subtracting 1

$5x = 10$

$x = \frac{10}{5}$ *Step 2:* Undo '× 5' by dividing by 5

$x = 2$ Check: $5 \times 2 + 1 = 11$

b **Method 1: The balancing method**

$\frac{5m}{2} = -3$

$\frac{5m}{2} \times 2 = -3 \times 2$ *Step 1:* Multiply both sides by 2

$5m = -6$

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$\frac{5m}{5} = \frac{-6}{5}$ *Step 2:* Divide both sides by 5

$m = -1\frac{1}{5}$ Check: $\frac{5 \times \left(-1\frac{1}{5}\right)}{2} = \frac{-6}{2} = -3$

Method 2: The backtracking method

$\frac{5m}{2} = -3$

$5m = -3 \times 2$ *Step 1:* Undo '÷ 2' by multiplying by 2

$5m = -6$

$m = \frac{-6}{5}$ *Step 2:* Undo '× 5' by dividing by 5

$m = -1\frac{1}{5}$ Check: $\frac{5 \times \left(-1\frac{1}{5}\right)}{2} = \frac{-6}{2} = -3$

c **Method 1: The balancing method**

$\frac{w}{3} - 6 = 5$

$\frac{w}{3} - 6 + 6 = 5 + 6$ *Step 1:* Add 6 to both sides

$\frac{w}{3} = 11$

$\frac{w}{3} \times 3 = 11 \times 3$ *Step 2:* Multiply both sides by 3

$w = 33$ Check: $\frac{33}{3} - 6 = 11 - 6 = 5$

Method 2: The backtracking method

$\frac{w}{3} - 6 = 5$

$\frac{w}{3} = 5 + 6$ *Step 1:* Undo '– 6' by adding 6

$\frac{w}{3} = 11$

$w = 11 \times 3$ *Step 2:* Undo '÷ 3' by multiplying by 3

$w = 33$ Check: $\frac{33}{3} - 6 = 11 - 6 = 5$

7.01

EXERCISE 7.01 ANSWERS ON P. 579

Two-step equations U F R

1 Solve each equation. Use substitution to check your solutions. R

EXAMPLE 1

a	$3x + 2 = 8$	**b**	$4p - 3 = 13$	**c**	$2w + 7 = 16$
d	$9k + 3 = 12k$	**e**	$8x = x - 3$	**f**	$10 + 5m = 20m$
g	$4 - 2y = 12$	**h**	$3 - 7x = 10$	**i**	$22 + 7h = 36$
j	$1.2 - 5q = 7.2$	**k**	$-12 - 8w = -2w$	**l**	$-6 - 4d = -20$

2 Solve $4x + 6 = 26$. Select the correct answer **A**, **B**, **C** or **D**.

A $x = 3$ **B** $x = 5$ **C** $x = 6$ **D** $x = 8$

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3 Solve each equation. Use substitution to check your solutions. R

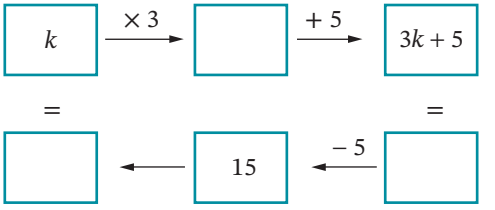
a $\frac{2n}{3}=4$ **b** $\frac{5x}{2}=7$ **c** $\frac{6a}{7}=8$

d $\frac{-2y}{9}=-8$ **e** $\frac{-5n}{3}=-9$ **f** $\frac{-6p}{11}=-4$

g $\frac{x-1}{3}=7$ **h** $\frac{n+6}{2}=8$ **i** $2=\frac{d+9}{5}$

j $\frac{a-3}{4}=-3$ **k** $\frac{n+3}{-5}=-1$ **l** $\frac{3-x}{-4}=2$

4 Copy and complete this flowchart to solve $3k + 5 = 20$ using the backtracking method. R

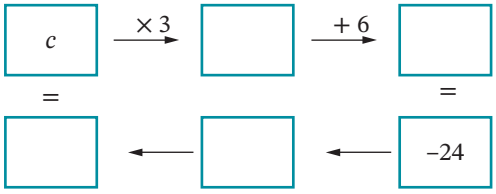


This is what has been done to k.

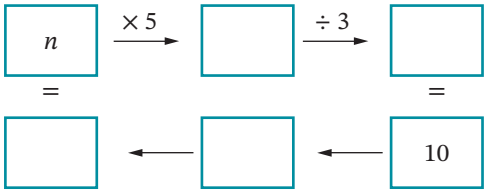
Undoing what has been done.

5 Copy and complete each flowchart to solve the equation using the backtracking method. R

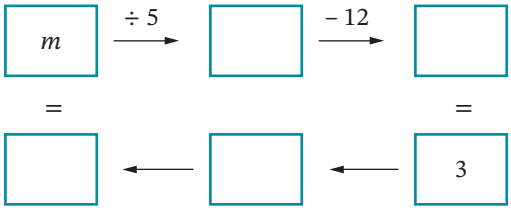
a $3c + 6 = -24$



b $\frac{5n}{3}=10$



c $\frac{m}{5} - 12 = 3$



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6 Solve each equation. R

a $\frac{w}{2}+3=4$ **b** $\frac{a}{6}-1=7$ **c** $\frac{m}{3}-12=8$

d $2-\frac{a}{3}=3$ **e** $1-\frac{x}{8}=-1$ **f** $4-\frac{a}{2}=-3$

7 Which one of these equations has the solution $x = -3$? Select **A**, **B**, **C** or **D**.

A $\frac{3x}{2}=3$ **B** $\frac{x-1}{2}=-1$ **C** $\frac{x}{3}-1=0$ **D** $2x+1=-5$

8 Solve each equation. R

a $\frac{x+8}{6}=3$ **b** $\frac{w}{7}+10=10$ **c** $\frac{2y}{11}=-6$

d $13+\frac{k}{4}=10$ **e** $\frac{x-3}{-4}=12$ **f** $-6-14d=-20$

9 What is the solution to $\frac{k}{10}-5=3$? Select **A**, **B**, **C** or **D**.

A $k = 80$ **B** $k=\frac{4}{5}$ **C** $k = -12$ **D** $k = -20$

Equations with variables on both sides 7.02

For equations with variables on both sides, such as $3x + 4 = 2x + 7$, we can only use the **balancing method**, not the backtracking method.



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Puzzle
Equations with unknowns on both sides

Video
Equations with variables on both sides

Equations with variables on both sides

For **equations with variables on both sides**, perform operations on both sides to move:

- all the variables onto one side of the equation
- all the numbers onto the other side of the equation.

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Example 2

Solve each equation.

a $7x + 7 = 2x + 2$ **b** $9 - 6y = 10 - 2y$

Solution

a

$7x + 7 = 2x + 2$

$7x - 2x + 7 = 2x - 2x + 2$ Subtracting $2x$ from both sides

$5x + 7 = 2$

$5x + 7 - 7 = 2 - 7$ Subtracting 7 from both sides

$5x = -5$

$\frac{5x}{5} = \frac{-5}{5}$ Dividing both sides by 5

$x = -1$

Check:

LHS $= 7 \times (-1) + 7 = 0$

RHS $= 2 \times (-1) + 2 = 0$

LHS = RHS.

LHS = left-hand side,
RHS = right-hand side.

b

$9 - 6y = 10 - 2y$

$9 - 6y + 2y = 10 - 2y + 2y$ Adding $2y$ to both sides

$9 - 4y = 10$

$9 - 4y - 9 = 10 - 9$ Subtracting 9 from both sides

$-4y = 1$

$\frac{-4y}{-4} = \frac{1}{-4}$ Dividing both sides by (-4)

$y = -\frac{1}{4}$

Check:

LHS $= 9 - 6 \times \left(-\frac{1}{4}\right) = 10\frac{1}{2}$

RHS $= 10 - 2 \times \left(-\frac{1}{4}\right) = 10\frac{1}{2}$

LHS = RHS.

EXERCISE 7.02 ANSWERS ON P. 579

Equations with variables on both sides U F R

1 Solve each equation and check your solutions. R

EXAMPLE 2

a	$5w + 3 = 2w + 21$	**b**	$2q - 10 = q - 4$	**c**	$13x + 1 = 8x + 26$
d	$12n + 3 = 5n - 11$	**e**	$8y - 10 = 10y - 30$	**f**	$3m - 2 = 10 - 3m$
g	$9 - 2a = a - 9$	**h**	$9 - 2x = 18 + 7x$	**i**	$12y + 6 = 6 + 9y$
j	$-12 - 10u = -20 - 18u$	**k**	$15 - 7x = 22 - 3x$	**l**	$10 - 6x = -15 - 11x$

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2 For each equation, select the correct solution **A**, **B**, **C** or **D**.

a $6x - 1 = 2x + 11$

A $x = 12$ **B** $x = 3$ **C** $x = 0$ **D** $x = 2.5$

b $11 - 4p = 2p + 2$

A $p = 6.5$ **B** $p = 2$ **C** $p = 1.5$ **D** $p = 3$

3 Solve each equation. R

a $7w + 15 = w + 3$ **b** $10 - 3t = 16 + t$ **c** $4a + 2 = 10 - 4a$

d $50 + 7y = 20 - 3y$ **e** $8y - 2 = 10y + 1$ **f** $9y + 3 = 9 - y$

g $9 - t = 7t - 2$ **h** $5y + 2 = 17 - y$ **i** $25 - 12k = 15 - 6k$

4 Solve $-3n - 8 = -7n - 12$. Select **A**, **B**, **C** or **D**.

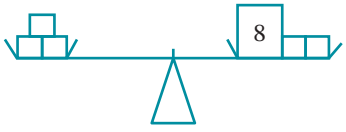
A $n = 5$ **B** $n = 2$ **C** $n = -1$ **D** $n = -0.4$

5 Each diagram below shows a balanced scale that represents an equation. Let each blank box be x. For each scale: R

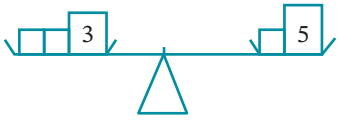
i write the equation that the scale represents and solve the equation

ii check your answer by substitution to see if the scale remains balanced.

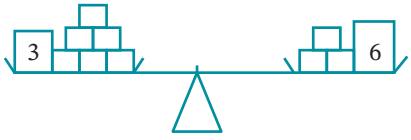
a



b



c



d



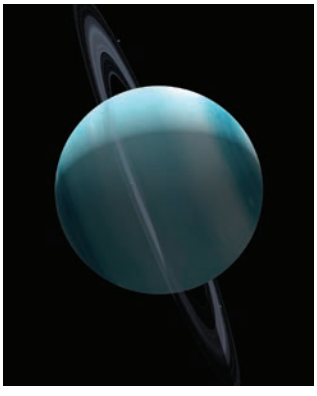
Did you know?

Uranus and Neptune

In 1781, British astronomer William Herschel discovered the planet Uranus (pictured). At that time, it was the farthest planet known in our solar system. However, astronomers found that Uranus' orbit around the Sun did not follow the expected path.

Working separately, mathematicians John Couch Adams of England and Urbain Leverrier of France both predicted that this different orbit was caused by an unknown planet. They calculated the position of this undiscovered planet using a number of equations.

In 1846, a German astronomer called Johann Galle located this planet and named it Neptune.



Science Photo Library/Claus Lunau

How long is one day on Uranus?

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7.03 Equations with brackets

Videos
Solving equations

Equations with brackets

Worksheets
Equations 2

Checking solutions

Puzzles
Equations

Equations order activity

Equations with brackets

For **equations with brackets (grouping symbols)**, expand the expressions and then solve as usual.

Example 3

Solve each equation.

a $3(a + 7) = 6$

b $9(m - 5) = 7(m + 1)$

Solution

a

$3(a + 7) = 6$

$3a + 21 = 6$ — Can you think of another way to solve this equation without expanding?

Expanding the expression to make it a 2-step equation

$3a + 21 - 21 = 6 - 21$

Subtract 21 from both sides

$3a = -15$

$\frac{3a}{3} = \frac{-15}{3}$

Divide both sides by 3

$a = -5$

Check: $3(-5 + 7) = 3 \times 2 = 6$

b

$9(m - 5) = 7(m + 1)$

$9m - 45 = 7m + 7$

Expanding brackets on both sides

$9m - 7m - 45 = 7m - 7m + 7$

Subtracting $7m$ from both sides

$2m - 45 = 7$

$2m - 45 + 45 = 7 + 45$

Adding 45 to both sides

$2m = 52$

$\frac{2m}{2} = \frac{52}{2}$

Dividing both sides by 2

$m = 26$

Check:

LHS $= 9 \times (26 - 5) = 9 \times 21 = 189$

RHS $= 7 \times (26 + 1) = 7 \times 27 = 189$

LHS = RHS.

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EXERCISE 7.03 ANSWERS ON P. 579

Equations with brackets U F R C

EXAMPLE 3

7.03

1 Solve each equation. R

a $2(m + 3) = 8$ **b** $3(x + 1) = 9$ **c** $5(y - 2) = 15y$

d $35 = 7(k + 1)$ **e** $4(3 - a) = 16$ **f** $11 = 9(1 + 2p)$

g $3h = 4(h + 6)$ **h** $6(m - 10) = -6$ **i** $8u = 11(u - 3)$

j $27 = 7(2y + 1)$ **k** $5(2 + 3p) = -8$ **l** $22x = 9(4x - 3)$

2 In which line has an error been made in solving $5(x - 3) = 25$? Select **A**, **B**, **C** or **D**.

$5(x - 3) = 25$

$5x - 8 = 25$ **A**

$5x - 8 + 8 = 25 + 8$ **B**

$5x = 33$ **C**

$x = \frac{33}{5}$ **D**

$= 6\frac{3}{5}$

3 Show that $k = 5$ is the solution to $12(k - 1) = 48$. R C

4 Show that $a = 6$ is the solution to $10 + a = 2(2 + a)$. R C

5 Solve each equation. R

a $8(m + 2) = 5(m + 5)$ **b** $2(y - 3) = 4(y - 5)$ **c** $3(2 + x) = 4(1 + x)$

d $5(p + 2) = 3(6 + p)$ **e** $5n + 6 = 2(2n + 1)$ **f** $2(4 - 3x) = 4(7 - 3x)$

g $4(3w - 1) = 5(4 + 3w)$ **h** $-2(x + 1) = 16 - 5x$ **i** $-8y - 5 = 5(2y - 3)$

6 Show that the solution to $5(2m - 2) = 6(m + 1)$ is $m = 4$. R C

7 Solve each equation. R

a $5(m + 6) + 10 = 3(m + 2) + 20$ **b** $3(y + 2) - 10 = 2(y - 1) + 5$

c $7y + 2(y + 5) = 4(y - 10)$ **d** $3x + 4(5 + x) = 6(2 + x) + 20$

e $5y + 2(y - 3) = 4y + 2(2y + 10)$ **f** $11 - 2(5 + y) = 4(3 + y) - 1$

g $8 - 3(1 - m) = 5(m + 3) + 4$ **h** $12 - 7(2y - 5) = 6 - 15(2 - 5y)$

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Investigation

Make your own equation

Here are 2 equations that have the same solution, $x = 6$:

$5x - 1 = 23 + x$ and $\frac{3x+12}{10} = 3$.

1 For each solution below, make up 3 equations that have that solution.

a $x = 6$ **b** $x = \frac{1}{4}$ **c** $x = 12$

d $x = 1\frac{1}{2}$ **e** $x = 0$ **f** $x = -2$

2 Compare your answers with those of other students. Check that each equation is correct.

7.04 Equation problems

Worksheets
Word problems with equations

Angle problems with algebra

Puzzle
Writing and solving equations

Steps in solving word problems using equations

1 Read the problem carefully and determine what needs to be found: 'What is the question?'

2 Use a variable to represent the unknown quantity.

3 Write the problem as an equation.

4 Solve the equation.

5 Answer the problem.

Example 4

When three-quarters of a number is decreased by 8, the result is 46. What is the number?

Solution

Let the number be x.

$\frac{3x}{4} - 8 = 46$ Translating from words to algebra

$\frac{3x}{4} - 8 + 8 = 46 + 8$ Adding 8 to both sides

$\frac{3x}{4} = 54$

$\frac{3x}{4} \times 4 = 54 \times 4$ Multiplying both sides by 4

$3x = 216$

$\frac{3x}{3} = \frac{216}{3}$ Dividing both sides by 3

$x = 72$ Check: $\frac{3 \times 72}{4} - 8 = 54 - 8 = 46$

The number is 72.

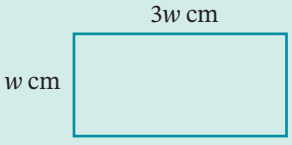
Example 5

A rectangle is 3 times as long as it is wide. If its perimeter is 60 cm, find its dimensions.

Solution

Let the width of the rectangle be w cm. Then the length is $3w$ cm.

The perimeter is $w + 3w + w + 3w$ and this is given as 60.



$w + 3w + w + 3w = 60$

$8w = 60$

$w = 7.5$

∴ The width of the rectangle is 7.5 cm and the length is $3 \times 7.5 = 22.5$ cm.

Check: The perimeter of a rectangle with dimensions 7.5 cm and 22.5 cm is $7.5 + 22.5 + 7.5 + 22.5 = 60$ cm.

Example 6

The sum of 3 consecutive numbers is 150.
Find the numbers.

Consecutive numbers follow each other in order, such as 7, 8, 9.

Solution

Let the first number be x.

The next number is $x + 1$ and the third number is $x + 2$.

Their sum is $x + (x + 1) + (x + 2)$ and this is given as 150.

$x + x + 1 + x + 2 = 150$

$3x + 3 = 150$

$3x = 147$

$x = \frac{147}{3}$

$= 49$

∴ The consecutive numbers are 49, 50 and 51.

Check: $49 + 50 + 51 = 150$.

Example 7

Ryan is 6 years older than his sister Kelsey.
Their mother is 3 times Ryan's age.

a If the sum of the 3 ages is 59, write a simplified equation to find Ryan's age.

b Solve the equation and find each person's age.



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Videos
Equation problems 1
Equation problems 2

Presentation
Applying linear equations

Solution

a Let x = Ryan's age.

Kelsey's age is $x - 6$. Kelsey is 6 years younger than Ryan

The mother's age is $3x$.

$x + (x - 6) + 3x = 59$

$5x - 6 = 59$ Simplify the equation

b $5x - 6 = 59$

$5x = 65$

$x = 13$

Ryan is 13 years old.

Kelsey is $13 - 6 = 7$ years old.

Their mother is $3 \times 13 = 39$ years old.

Check: $13 + 7 + 39 = 59$

EXERCISE 7.04 ANSWERS ON P. 579

Equation problems U F PS R C

EXAMPLE 4

1 When 7 is subtracted from 4 times a certain number, the answer is 37. What is the number? PS R C

2 If 15 more than a number is 3 more than double the number, what is the number?

3 Two-thirds of a number is 16. What is the number?

4 When two-fifths of a number is increased by 15, the result is 27. What is the number?

EXAMPLE 5

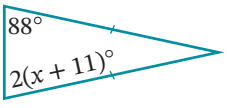
5 A rectangle is 4 times as long as it is wide. The perimeter of the rectangle is 100 cm. Find the dimensions of the rectangle. PS R C

6 The length of a rectangle is 7 cm longer than its width.

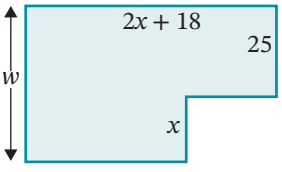
a Let w be the width of the rectangle. Write an equation for w if the perimeter of the rectangle is 94 cm.

b Solve the equation and find the dimensions of the rectangle.

7 Find the value of x in this triangle.



8 The block of land shown has a perimeter of 165 m. Find the width (w) of the block. PS R C



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9 Brad had to walk, run and then catch a bus to get to the railway station. The distance to the railway station was 5.8 km, the bus ride was 3.5 km and Brad walked 4 times as far as he ran. How far did Brad run? Give your answer in metres. PS R C

10 The sum of 2 consecutive numbers is 87. Find the numbers. PS R C

EXAMPLE 6

11 The sum of 3 consecutive numbers is 87. Find the numbers.

12 The sum of 3 consecutive even integers is 168. What are the 3 integers?

7.04

13 The sum of 3 consecutive *odd* integers is 75. Find the integers.

14 Dean's father, Franco, is 5 times Dean's age. Dean is 8 years older than his sister, Helen. The sum of all their ages is 62 years. How old is each person? (*Hint:* Let Dean's age be x.) PS R C

EXAMPLE 7

15 Bill is 3 times as old as his daughter, Rebekah, who is 6 years younger than her brother, Tyrone. How old is Rebekah if the sum of their 3 ages is 76 years?

16 In my money box, I have only \$1 and \$2 coins. I have 240 coins in total, worth \$318. How many \$2 coins do I have in the money box?

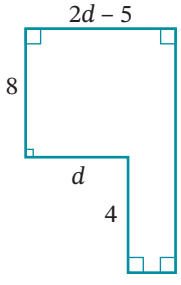
17 Aerin bought 4 ice creams and received \$2.80 change from his \$20 note. How much did each ice cream cost?

18 Ashvin is twice as tall as his little sister, Sureti, and 30 cm shorter than his father. Their combined height is 3.8 m. Find the height of each person.

19 The length of a rectangle is 9.5 cm longer than its width. The perimeter of the rectangle is 87 cm. Find the dimensions of the rectangle.

20 Janine is 6 years younger than her husband, Vila, who is 3 times the age of their son Brett. Brett is 5 years older than his sister Amanda. The sum of all their ages is 125 years. How old is each person?

21 A house is constructed in the shape shown. If the house has a floor area of 54m^2, what is the value of d? PS R C



22 Ashleigh only collects 20-cent and 50-cent coins. Last week she counted 420 coins to the value of \$156. How many 50-cent coins did Ashleigh have?

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Quiz
Mental skills 7A

Mental skills 7A: Maths without calculators ANSWERS ON P. 580

Fraction of a quantity

Learn these commonly used fractions and their decimal equivalents.

Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{2}{5}$
Decimal	0.5	0.25	0.125	0.75	0.2	0.1	0.05	0.4

Now we will use them to find a fraction or decimal of a quantity.

1 Study each example.

a $\frac{1}{4} \times 72 = 72 \div 4$
$= 18$

b $\frac{2}{5} \times 40 = (\frac{1}{5} \times 40) \times 2$
$= 8 \times 2$
$= 16$

c $\frac{3}{4} \times 32 = (\frac{1}{4} \times 32) \times 3$
$= 8 \times 3$
$= 24$

d $0.5 \times 66 = \frac{1}{2} \times 66$
$= 33$

e $0.05 \times 80 = \frac{1}{20} \times 80$
$= 4$

f $0.125 \times 56 = \frac{1}{8} \times 56$
$= 7$

2 Now simplify each expression.

a $\frac{1}{2} \times 28$ **b** $\frac{1}{4} \times 36$ **c** $\frac{1}{10} \times 70$ **d** $\frac{1}{8} \times 64$

e $\frac{1}{5} \times 15$ **f** $\frac{1}{10} \times 80$ **g** $\frac{2}{5} \times 25$ **h** $\frac{1}{20} \times 100$

i $\frac{3}{4} \times 44$ **j** $\frac{1}{8} \times 40$ **k** 0.25×60 **l** 0.4×45

m 0.1×260 **n** 0.125×48 **o** 0.75×48 **p** 0.05×120

q 0.2×70 **r** 0.5×320 **s** 0.25×56 **t** 0.125×16

Investigation

Solving $x^2 = c$

1 $x^2 = 25$ has 2 solutions. What are they?

2 What are the possible solutions for each of the following?

a $x^2 = 9$ **b** $x^2 = 49$ **c** $x^2 = 100$

3 What is the inverse operation of 'squaring'?

4 Study this example:

$x^2 = 81$

$x = \pm\sqrt{81}$ which means $\sqrt{81}$ or $-\sqrt{81}$

$x = \pm 9$ which means 9 or –9

Check: When $x = 9$, $x^2 = 9^2 = 81$

When $x = -9$, $x^2 = (-9)^2 = 81$

Now use the same method to solve each equation and check your answers.

a $m^2 = 1$ **b** $k^2 = 64$

5 How many solutions does each quadratic equation have?

a $m^2 = 1$ **b** $k^2 = 64$ **c** $x^2 = 81$

6 Do the following quadratic equations have solutions? (Give reasons for your answers.)

a $w^2 = -1$ **b** $y^2 = -64$ **c** $h^2 = -81$

7 Write an example of a quadratic equation that has only *one* solution.

Simple quadratic equations $ax^2 = c$ 7.05

An equation in which the highest power of the variable is 2 is called a **quadratic equation**, for example, $x^2 = 5$, $3m^2 + 7 = 10$, $d^2 - 4 = 0$ and $4t^2 - 3t = 8$.

The quadratic equation $x^2 = c$

The quadratic equation $x^2 = c$ (where c is a positive number) has 2 solutions:

$x = \pm\sqrt{c}$ (which means $x = \sqrt{c}$ and $x = -\sqrt{c}$).

Example 8

Solve each quadratic equation.

a $y^2 = 16$ **b** $p^2 = 65$ **c** $5a^2 = 245$



Video
Simple quadratic equations $ax^2 = c$

Solution

a $y^2 = 16$

$y = \pm\sqrt{16}$ Finding the square root of both sides

$= \pm 4$

b $p^2 = 65$

$p = \pm\sqrt{65}$ 65 is not a square number, so leave the answer as a surd. Finding the square root of both sides

c $5a^2 = 245$

$a^2 = \frac{245}{5}$ Dividing both sides by 5

$a^2 = 49$

$a = \pm\sqrt{49}$

$= \pm 7$

Example 9

Solve each quadratic equation, writing the solution correct to one decimal place.

a $4x^2 = 600$

b $\frac{3h^2}{5} = 26$

Solution

a
$$4x^2 = 600$$
$$x^2 = \frac{600}{4}$$
$$x^2 = 125$$
$$x^2 = \pm\sqrt{125}$$
$$= \pm 11.1803...$$
$$\approx \pm 11.2$$

b
$$\frac{3h^2}{5} = 26$$
$$3h^2 = 26 \times 5$$
$$3h^2 = 130$$
$$h^2 = \frac{130}{3}$$
$$h^2 = 43\frac{1}{3}$$
$$h = \pm\sqrt{43\frac{1}{3}}$$
$$= \pm 6.5828...$$
$$\approx \pm 6.6$$

EXERCISE 7.05 ANSWERS ON P. 580

Simple quadratic equations $ax^2 = c$ U F R C

EXAMPLE 8

1 Solve each quadratic equation, writing the solutions as surds if necessary.

a	$m^2 = 144$	**b**	$x^2 = 400$	**c**	$y^2 = 225$
d	$k^2 = 59$	**e**	$y^2 = 10$	**f**	$w^2 = 36$
g	$8x^2 = 200$	**h**	$9t^2 = 81$	**i**	$\frac{a^2}{2} = 8$
j	$5k^2 = 40$	**k**	$3w^2 = 30$	**l**	$2d^2 = 288$
m	$\frac{k^2}{2} = 8$	**n**	$\frac{w^2}{10} = 7$	**o**	$4x^2 = 1$
p	$\frac{m^2}{3} = 27$	**q**	$8y^2 = 40$	**r**	$2p^2 + 3 = 21$
s	$3k^2 = 48$	**t**	$\frac{y^2}{5} - 2 = 9$	**u**	$6x^2 = 42$

EXAMPLE 9

2 Solve each equation, writing the solution correct to one decimal place.

a	$m^2 = 20$	**b**	$b^2 = 17$	**c**	$v^2 = 6$
d	$2p^2 = 35$	**e**	$9k^2 = 63$	**f**	$\frac{x^2}{5} = 8$
g	$\frac{k^2}{16} = 6$	**h**	$\frac{7u^2}{10} = 2$	**i**	$6y^2 = 84$
j	$\frac{3w^2}{4} = 20$	**k**	$a^2 + 11 = 28$	**l**	$2y^2 - 14 = 63$

3 Explain why the quadratic equation $k^2 + 25 = 0$ has no solutions. R C

4 State which of these quadratic equations has no solutions. Give reasons. R C

a	$x^2 = -9$	**b**	$2k^2 + 5 = 9$	**c**	$3m^2 + 8 = 4$
d	$\frac{9w^2}{2} - 1 = 1$	**e**	$4 + \frac{d^2}{3} = 8$	**f**	$\frac{5a^2}{2} + 3 = 2$

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Equations and formulas 7.06

A **formula** is an algebraic equation that shows a relationship between variables. For example, the formula for the area of a circle is $A = \pi r^2$, where A is the area and r is the radius of the circle (π is a constant). Because the formula is for the area, A is called the **subject** of the formula and it is the variable on its own on the left side of the '=' sign.

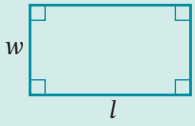
Worksheet
Working with formulas

Videos
Equations and formulas 1
Equations and formulas 2

Example 10

The formula for the perimeter (P) of a rectangle of length l and width w is given by

$P = 2(l + w)$.



Use the formula to find:

a the perimeter of a rectangle with length 20 cm and width 9 cm

b the width of a rectangle if its length is 12 m and its perimeter is 70 m

c the length of a rectangle if its width is 42 cm and its perimeter is 1.8 m.

Solution

a $l = 20, w = 9$:

$P = 2(l + w)$

$= 2(20 + 9)$

$= 2 \times 29$

$= 58$

$\therefore$ The perimeter is 58 cm.

b $l = 12, P = 70$:

$P = 2(l + w)$

$70 = 2(12 + w)$

$70 = 24 + 2w$

$46 = 2w$

$w = \frac{46}{2}$

$= 23$

$\therefore$ The width is 23 cm.

c $w = 42, P = 1.8 \text{ m} = 180 \text{ cm}$ since w is given in cm

$P = 2(l + w)$

$180 = 2(l + 42)$

$180 = 2l + 84$

$96 = 2l$

$l = \frac{96}{2}$

$= 48$

$\therefore$ The length is 48 cm.

Example 11

The cost of hiring a sound system for a party is $250 plus $75 per hour. The cost can be represented by the formula $C = 250 + 75h$, where C is the total cost (in dollars), and h is the number of hours.

a Find the cost of hiring the sound system for 4 hours.

b A family is willing to spend $750 for hiring the sound system. What is the maximum number of *whole* rental hours that the family can afford?



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Solution

a $h = 4$:

$C = 250 + 75h$

$= 250 + 75 \times 4$

$= 550$

$\therefore$ The cost is $550.

b $C = 750$:

$C = 250 + 75h$

$750 = 250 + 75h$

$500 = 75h$

$h = \frac{500}{75}$

$= 6\frac{2}{3}$

$\therefore$ The maximum number of whole hours is 6.

EXERCISE 7.06 ANSWERS ON P. 580

Equations and formulas U F PS R

1 Given the formula $y = 5x + c$, find:

a y if $x = 5$ and $c = 3$

b y if $x = -1$ and $c = 16$

c c if $y = 40$ and $x = 3$

d c if $y = -6$ and $x = -1$

e x if $y = 27$ and $c = 12$

f x if $y = 64$ and $c = -16$

EXAMPLE 10

2 A temperature in degrees Celsius (°C) can be converted to degrees Fahrenheit (°F) using the formula $F = \frac{9C}{5} + 32$. Convert each temperature to °F.

a 35°C **b** −10°C **c** 16°C



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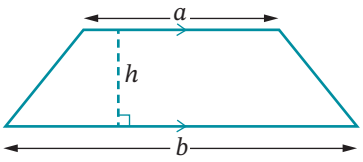
3 Use the formula in question **2** to convert each temperature to °C, correct to one decimal place. PS R

a 100°F **b** −45°F **c** 78°F

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4 The formula $A = \frac{1}{2}(a + b)h$ is used to find the area of a trapezium, where A is the area, a and b are the lengths of the parallel sides and h is the perpendicular height between them.



Use the formula to find:

a the area of a trapezium with height 6 cm and parallel sides of length 9 cm and 15 cm

b the height of a trapezium if its area is 420 cm^2 and it has parallel sides of length 22 cm and 20 cm

c the length of one side of a trapezium if its parallel side is 20.5 m, its area is 318 m^2 and its height is 12 m.

5 Find the value of S in the formula $M = \frac{kS}{5}$ if $M = 12.6$ and $k = 3.15$. Select **A**, **B**, **C** or **D**.

A 60 **B** 7.938 **C** 20 **D** 0.8

6 The cost C (in dollars) of hiring a limousine is given by the formula $C = 2500 + 150h$, where h is the number of hours of hire. Find: PS R

a the cost of hiring a limousine for 4 hours

b the cost of hiring a limousine for 2 days

c the number of hours for which you could hire a limousine for $4000

d the maximum number of whole hours a limousine could be hired for $5250

7 The profit, $\$P$, made by an online store selling games is given by $P = 35x - 750$, where x represents the number of games sold. Find: PS R

a the profit made when 230 games are sold

b the number of games sold if the profit is $18 010.

8 A catering company charges $\$C$ for a function with P people using the formula $C = 1500 + 75.5P$. PS R

a How much does the company charge for a function with 10 guests?

b Find the cost of catering for a group of 60 people.

c Diane has $10 000 to spend on catering for her wedding reception. What is the maximum number of people she can invite?

9 The perimeter of a rectangle is given by the formula $P = 2(l + w)$, where l is the rectangle's length and w is its width. What is the length of a rectangle with a width of 4 and a perimeter of 52? Select **A**, **B**, **C** or **D**.

A 9 **B** 12 **C** 18 **D** 22

10 The temperature T (in °C) of a hot liquid as it cools is given by the formula $T = 100 - 17.5h$, where h is the number of hours it has been cooling. Find: PS R

a the temperature of the liquid after 2 hours

b the temperature of the liquid after 30 minutes

c the number of hours it takes for the temperature of the liquid to reach 30°C.

7.06

EXAMPLE 11

Foundation Standard Complex

11 Archeologists use the formula $H = 2.52t + 75.8$ to estimate the height, H cm, of a man when the tibia (shin bone) length, t cm, is measurable. PS R

a A male tibia 42 cm long was found intact. Estimate the height of the male, correct to the nearest centimetre.

b Estimate, to the nearest centimetre, the length of the tibia of a male whose height is 174 cm.

Quiz
Mental skills 7B

Mental skills 7B: Maths without calculators ANSWERS ON P. 580

Percentage of a quantity

Learn these commonly used percentages and their fraction equivalents.

Percentage	50%	25%	12.5%	75%	20%	10%	$33\frac{1}{3}\%$	$66\frac{2}{3}\%$
Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{3}$	$\frac{2}{3}$

Now we will use them to find a percentage of a quantity.

1 Study each example.

a $20\% \times 25 = \frac{1}{5} \times 25$
$= 5$

b $50\% \times 120 = \frac{1}{2} \times 120$
$= 60$

c $12.5\% \times 32 = \frac{1}{8} \times 32$
$= 4$

d $75\% \times 56 = \frac{3}{4} \times 60$
$= (\frac{1}{4} \times 60) \times 3$
$= 15 \times 3$
$= 45$

e $33\frac{1}{3}\% \times 27 = \frac{1}{3} \times 27$
$= 9$

f $66\frac{2}{3}\% \times 60 = \frac{2}{3} \times 60$
$= (\frac{1}{3} \times 60) \times 2$
$= 20 \times 2$
$= 40$

2 Now simplify each expression.

a $25\% \times 44$ **b** $33\frac{1}{3}\% \times 120$ **c** $20\% \times 35$ **d** $66\frac{2}{3}\% \times 36$

e $10\% \times 230$ **f** $12\frac{1}{2}\% \times 48$ **g** $50\% \times 86$ **h** $20\% \times 400$

i $75\% \times 24$ **j** $33\frac{1}{3}\% \times 45$ **k** $25\% \times 160$ **l** $10\% \times 650$

m $12.5\% \times 88$ **n** $66\frac{2}{3}\% \times 21$ **o** $20\% \times 60$ **p** $75\% \times 180$

Power plus ANSWERS ON P. 580

1 Solve each equation.

a $11y = 16 - 3(10 - 4y)$

b $5(2s - 10) + 3(2s + 1) = 2(5 - s)$

c $17 - 5(3t - 2) = 12 - 10(9 - t)$

d $6(4 - x) - 2(x - 1) = 3(x + 10) - 5(x - 4)$

e $\frac{-3(2m+5)}{4} = 6$

f $\frac{d-3}{4} + \frac{2}{5} = 4$

g $\frac{5x}{2} - \frac{3x-1}{4} = 9$

h $\frac{2y-1}{4} = 5 + \frac{y+2}{3}$

i $\frac{5}{2y} = 14$

j $\frac{7}{r} = \frac{1}{r-3}$

k $\frac{y^2+8}{12} = 4\frac{2}{3}$

l $4m^2 - 3 = m^2 + 72$

2 Given $W = \frac{X-Y}{X+Y}$, find:

a W when $X = 15$ and $Y = 10$

b W when $X = 6$ and $Y = -12$

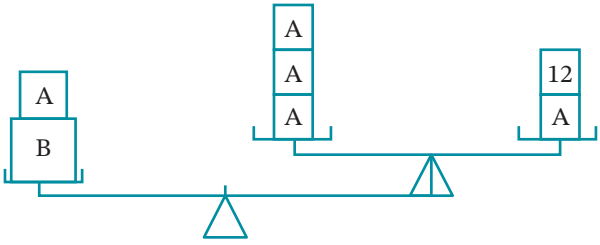
c X when $W = 25$ and $Y = 6$

d Y when $W = 5$ and $X = 1$

3 Karlie is a farmer who raises pigs and chickens. From a total of 42 animals, she can count 116 legs. Write an equation and solve it to find how many chickens she has.

4 Joe is twice as old as his daughter, Wendi. 10 years ago, he was 3 times as old as her. Write an equation and solve it to calculate how old Wendi is now.

5 Find the value of B in the balance scale below.



CHAPTER 7 REVIEW

Quiz
Language of maths 7

Puzzle
Equations crossword

Language of maths

backtracking method	balancing method	brackets
consecutive	equation	expand
flowchart	formula	inverse operation
LHS	lowest common multiple (LCM)	linear equation
quadratic equation	RHS	solution
solve	square root	subject
substitute	surd	undoing
unknown	variable	

1 Which method of solving equations involves using inverse operations on both sides of the equation?

2 What is the **subject** of the formula $A = \frac{1}{2}(a + b)h$?

3 What name is given to numbers that follow each other in order, such as 9, 10, 11?

4 Write an example of:

a a quadratic equation **b** a linear equation

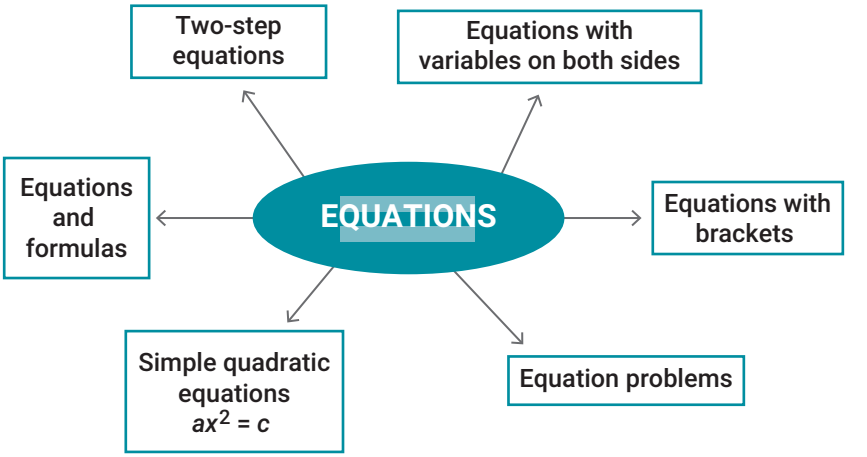
5 How many solutions does a linear equation have?

6 What does **LHS** stand for?

Topic summary

Worksheet
Mind map: Equations

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 7

ANSWERS ON P. 580

Quiz
Test yourself 7

1 Solve each equation. Check your solutions. 7.01

a $3x - 10 = 5$ **b** $4 + 2y = 21$ **c** $12y + 5 = 23$

d $\frac{m}{4} - 6 = 2$ **e** $\frac{x-5}{4} = 7$ **f** $\frac{5x}{3} = 10$

g $\frac{-2x}{5} = 3$ **h** $11 - 2a = 17$ **i** $20 + 4d = -6$

2 Solve each equation. Check your solutions. 7.02

a $2x + 5 = 3x + 4$ **b** $7 + 4x = x - 8$ **c** $5 - 6w = 3w - 7$

d $3x + 4 = 2x + 7$ **e** $5n - 3 = 2n - 15$ **f** $2d - 8 = -5d - 71$

g $4t = 12 - 4t$ **h** $8j - 17 = 10j$ **i** $6 - 3q = 8 - q$

3 Write an equation with x on both sides that has the solution $x = 3$.

4 Solve each equation. 7.03

a $2(w - 5) = 4$ **b** $3(1 + 4n) = 15$ **c** $5(1 - 3p) = 20$

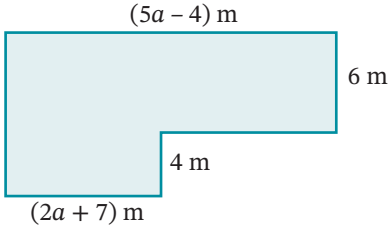
d $2(3 + x) = 5(x + 1)$ **e** $3(1 - y) = 4(2 - y)$ **f** $2(3 - 4x) = -(2x + 3)$

5 Solve each equation. 7.03

a $18 - 3a = 5a - 2(a - 6)$ **b** $11g + 2(g + 3) = 5(2g + 10)$

6 The length of a rectangle is 6 cm longer than it is wide. The perimeter of the rectangle is 76 cm. Find the dimensions of the rectangle. 7.04

7 The diagram shows the floor plan of a house. If the area of the floor is 99 m^2, find the value of a. 7.04



8 If 6 more than a number equals 5 more than double the number, what is the number? 7.04

9 Solve each quadratic equation. 7.05

a $d^2 = 64$ **b** $8p^2 = 288$ **c** $3z^2 = 105$

10 The body mass index (BMI) of an adult is $B = \frac{M}{h^2}$, where M is the mass in kilograms and h is the height in metres. 7.06

a Find the BMI of Ned who is 1.85 m tall and has a mass of 72 kg.

b Find the mass of a person with a BMI of 24, who is 2.1 m tall.

11 The cost, C, in dollars, of hiring a taxi is $C = 5 + 2.4d$, where d is the distance travelled in kilometres. Find: 7.06

a the cost of a taxi trip if the distance travelled is 15 km

b the distance travelled if the cost of a taxi trip was $78.20.

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PRACTICE SET 2

ANSWERS ON P. 580

5.07 **1** State the number of significant figures in each number.

a 1203 **b** 0.008 **c** 1.092 **d** 15 700

5.01 **2** Simplify each expression.

a $y^9 \times y$ **b** $3g^7 \times 4g^8$ **c** $2p^3q \times 9p^2q^2$

d $\frac{k^{18}}{k^6}$ **e** $\frac{x^{11}}{x^7}$ **f** $h^{10} \div h$

g $54k^7 \div 6k^3$ **h** $\frac{h^3y^7}{hy^4}$ **i** $12m^5n^8 \div 18mn^7$

5.04 **3** Simplify each expression.

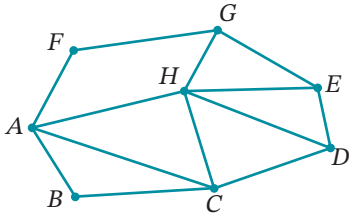
a $6y^0$ **b** $4^0 \times 3^0$ **c** $(4k)^0$ **d** $x \times (3y)^0$ **e** $2q^0 - (2q)^0$

4.04 **4** Evaluate each expression, correct to one decimal place.

a $16 \cos 55°$ **b** $8.7 \tan 13°$ **c** $\frac{24.5}{\sin 37°}$ **d** $\frac{40}{\cos 76.3°}$

PATH

5 This graph shows a road network between 8 towns.



6.03 **a** Find the degree of each vertex.

6.04 **b** State whether each walk is a trail or a path.

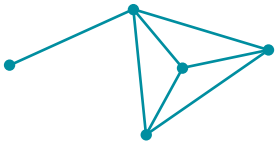
i *ACHGEHA* **ii** *HDEGFA*

iii *AHDCBA* **iv** *HDEGFAHC*

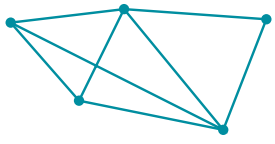
c Why is this graph a planar graph?

6.03 **6** Count the number of faces on each planar graph.

a



b



c



5.05 **7** Evaluate each expression, giving your answers in fraction form.

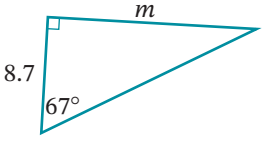
a 2^{-1} **b** 5^{-3} **c** 10^{-2}

4.07 **8** Solve $\tan\alpha = \frac{9}{13}$ for angle α, correct to the nearest minute. Select **A**, **B**, **C** or **D**.

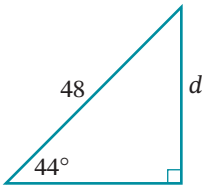
A 34° 69′ **B** 35° 10′ **C** 34° 41′ **D** 34° 42′

4.05 **9** Calculate the value of each variable, correct to 2 decimal places.

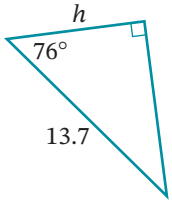
a



b



c



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10 Simplify each expression, giving the answer in index notation. 5.02

a $(2^3)^5$ **b** $(4^2)^6$ **c** $(3p^7)^2$ **d** $(2w^5)^3$

11 Express each number in decimal form. 5.08

a 1.54×10^7 **b** 3.2×10^{-3} **c** 9.837×10^{-1} **d** 2.4×10^4

12 Find A, correct to the nearest degree, if: 4.07

a $\sin A = \frac{4}{5}$ **b** $\tan A = 4.713$ **c** $\cos A = 0.5148$

13 Find θ, correct to the nearest minute, if: 4.07

a $\cos \theta = 0.21$ **b** $\sin \theta = \frac{9}{10}$ **c** $\tan \theta = 1.5$

14 Write each expression using a negative index. 5.05

a $\frac{1}{n}$ **b** $\frac{1}{2^3}$ **c** $\frac{1}{k^5}$ **d** $\frac{7}{y^2}$

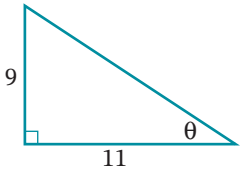
15 Write 7 280 000 in scientific notation. 5.08

16 Evaluate each expression, giving your answers in scientific notation, correct to 2 significant figures. 5.09

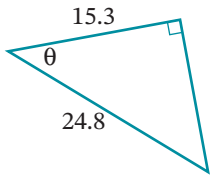
a $(7.08 \times 10^7) \times (3.8 \times 10^8)$ **b** $(4.65 \times 10^{-3}) \div (3.9 \times 10^7)$

17 Find θ, correct to the nearest degree. 4.07

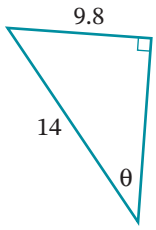
a



b



c



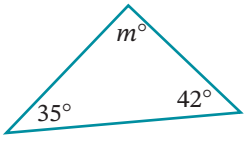
18 Solve each equation. 7.01

a $5x - 9 = 11$ **b** $1 - 4x = 10$ **c** $-3 = 7 + 8x$

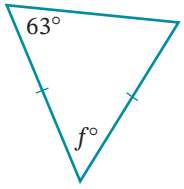
d $\frac{y+2}{3} = 5$ **e** $1 + \frac{a}{3} = 7$ **f** $8 = \frac{2x-1}{7}$

19 Find the value of each variable, giving reasons. 6.01

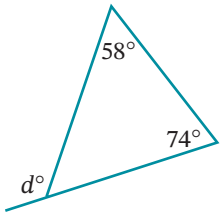
a



b



c



20 Solve $12 - 5k = 2k - 9$. Select the correct answer **A**, **B**, **C** or **D**. 7.02

A $k = 3$ **B** $k = 1$ **C** $k = \frac{3}{7}$ **D** $k = 7$

21 Select the larger number for each pair of numbers. 5.08

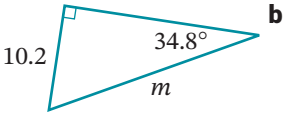
a 5.3×10^9 or 8.6×10^8 **b** 3.73×10^5 or 3.4×10^5

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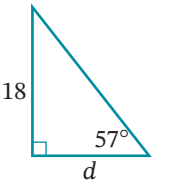
4.06

22 Find the value of each variable, correct to one decimal place.

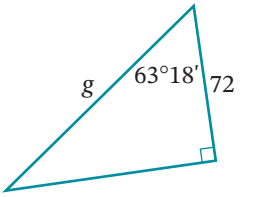
a



b



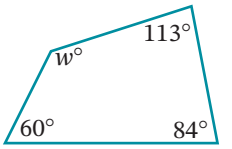
c



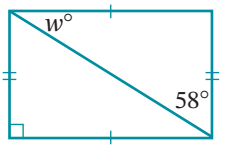
6.02

23 Find the value of w in each quadrilateral.

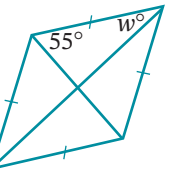
a



b



c



7.06

24 The volume of useable timber (in cubic metres) in a tree is calculated using the formula:

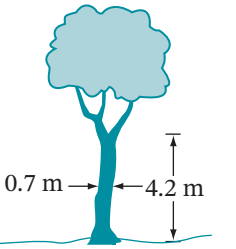
$$v = 0.5hd^2 + 10$$

where: d = the diameter of the tree

h = the height to the first branch

Calculate the volume of useable timber in this tree.

Select **A**, **B**, **C** or **D**.

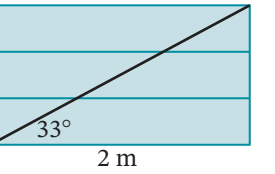


A 16.2 m³ **B** 11.5 m³

C 11.0 m³ **D** 18.8 m³

4.06

25 How long is the diagonal brace of this gate, correct to the nearest centimetre?



4.07

26 In $\triangle ABC$, $\angle A = 90°$, $AC = 12.7$ cm and $BC = 17.8$ cm. Find $\angle B$ correct to the nearest minute.

5.03

27 Simplify each expression.

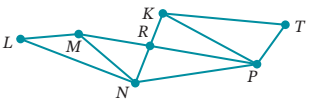
a $(2m^3n^2)^4$ **b** $\left(-\frac{2h}{5}\right)^2$ **c** $(7rt^2)^3$

PATH

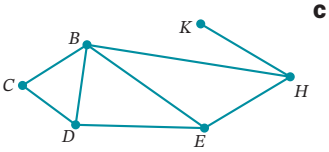
28 For each planar graph, count the number of vertices, edges and faces and verify Euler's formula $v - e + f = 2$.

6.04

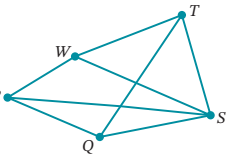
a



b



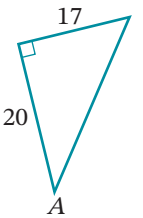
c



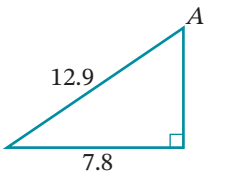
4.07

29 Find the size of angle A, correct to the nearest degree.

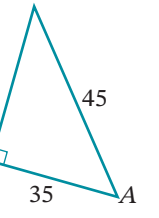
a



b



c



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30 A formula for cooking roast beef is $T = 40M + 25$, where T is the cooking time in minutes and M is the mass of beef in kilograms. 7.06

a How long will it take to cook a 5 kg piece of beef?

b What mass of beef has a cooking time of $2\frac{1}{2}$ hours?

31 A 3.5 m ladder leans against a wall, making an angle of 70° with the ground. 4.05

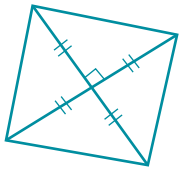
a Draw a diagram showing this information.

b Calculate, correct to the nearest 0.1 m:

i how far up the wall the ladder reaches

ii how far the foot of the ladder is from the base of the wall.

32 What type of quadrilateral is shown here? 6.02



33 Solve each equation. 7.03

a $3(p - 3) = 24$ **b** $3(4y - 1) + 5 = 2(9 - 2y)$

34 Round: 5.07

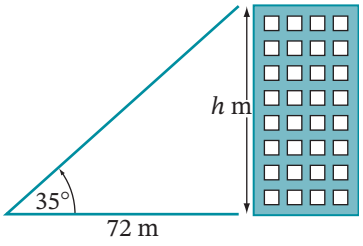
a 43 759 to 2 significant figures **b** 10.7614 to 3 significant figures

c 0.000 038 9 to one significant figure

35 **a** The sum of 4 consecutive numbers is 82. Find the 4 numbers. 7.04

b Sean collected \$1 and \$2 coins until he had 275 coins, having a total value of \$402. How many \$2 coins did Sean have?

36 Amit is at ground level 72 metres from the base of an apartment building. The angle to the top of the block is 35°. Find the height of the building, correct to the nearest metre. 4.05



37 $a = 4$ is the solution to *more than one* of the following equations. Select all correct answers **A**, **B**, **C** or **D**. 7.02

A $3a - 5 = 2a - 1$ **B** $2a + 3 = 11 - a$ **C** $\frac{a}{2} - 2 = 0$ **D** $16 - a = 3a$

38 Solve each equation. 7.05

a $5m^2 + 1 = 81$ **b** $\frac{x^2}{3} = 12$

39 The formula $A = \frac{1}{2}(a+b)h$ is used to find the area of the trapezium. Find the height if the area is 920 cm^2 and the 2 parallel sides are 25 cm and 21 cm. 7.06